

REVISED POC SCOPE

Intelligent Invoice Approval Automation System

Client Review Feedback Incorporated — v5

This is a complete, buildable POC — every item the client flagged as a gap is now baseline scope, not a future enhancement: duplicate-invoice detection, a full status lifecycle, role-based dashboards, confidence-tier automation, a notification engine, search, expanded audit logging, 5-role RBAC, invoice history timelines, and analytics. The AI pipeline and validation layer are expanded to reduce hallucination risk and enforce data integrity before anything reaches a human reviewer. Sections 15–20 give the architecture, schema, API contracts, stack, delivery plan, and acceptance criteria needed to build this end-to-end with nothing left ambiguous.

Net effect: broader scope than the original 8–9 week estimate. Revised timeline and budget are in Section 14 and Section 19.

1. Duplicate Invoice Detection — moved into MVP

On submission, the system checks Vendor + Invoice Number + Amount against existing records before allowing processing to continue.

- Match found → invoice is held with a "⚠ Possible Duplicate Invoice" flag instead of being auto-processed
- BA must explicitly confirm or dismiss the flag before the invoice can proceed
- Every duplicate flag and its resolution is written to the audit log

2. Invoice Status Lifecycle

Replaces the original three-state model (Pending / Approved / Rejected) with a full lifecycle that drives every dashboard and the invoice history timeline.

Received → AI Extracted → Awaiting BA Review → Awaiting Approval → Approved / Rejected → Sent to Finance → Completed

Rejected invoices exit the approval branch with a notification back to the BA; every other status advances linearly.

3. AI Processing Pipeline — expanded to reduce hallucination

The original two-stage OCR → LLM pipeline is replaced with a nine-stage pipeline. Each stage narrows what the model is asked to do, which is what actually reduces hallucination — not the LLM step itself.

Email → Virus Scan → OCR → Document Classification → Invoice Detection → AI Extraction → Validation Rules → Confidence Score → Human Review

4. Confidence Threshold Automation

Confidence	Behaviour
> 95%	Field auto-filled, no BA action required
70% – 95%	Field highlighted yellow — BA should verify
< 70%	Field highlighted red — BA must correct before submission

5. Validation Rules

Applied at the Validation Rules stage of the AI pipeline (Section 3) and again on BA submission.

- Invoice Number and Vendor are mandatory

- Due Date must be later than Invoice Date
- Tax must be ≥ 0
- Total must equal the sum of line items
- Currency code must be valid (ISO 4217)
- PO Number must match the agreed format
- Duplicate check (Section 1)
- Invoice Date cannot be in the future
- No mandatory fields left blank

6. Role-Based Dashboards

BA Dashboard

- Pending Reviews
- Today's Reviews
- Low Confidence Invoices
- Rejected
- Average Processing Time

Department Head Dashboard

- Pending Signatures
- Urgent Approvals
- Recently Approved

Finance Dashboard

- Approved Invoices
- Waiting for BMS Entry
- Completed

Admin Dashboard

- Invoices Today
- Extraction Accuracy
- OCR Failures
- Average Approval Time
- Users
- Storage Usage

7. Role-Based Access Control

Role	Access
Admin	Full system configuration, user management, all dashboards
BA	Review queue, correction interface, BA dashboard
Department Head	Approval, comment, signature, own dashboard
Finance	Signed documents, handoff package, Finance dashboard
Read-Only Auditor	View-only access to invoices, audit logs, and history — no edit rights

8. Notification Engine

Trigger	Recipient	Timing
Invoice extracted, awaiting review	BA	Immediate
Invoice submitted, awaiting signature	Department Head	Immediate
Approval pending	Department Head	Reminder at 24h and 48h
Invoice approved	Finance	Immediate
Invoice rejected	BA	Immediate

9. Search

Global search across Vendor, Invoice Number, PO Number, Amount, Status, Date, and Department. Flagged as a hard requirement — without it, the system becomes unusable past a few hundred invoices.

10. Audit Logs — expanded for Australian compliance

- Who opened / viewed each invoice
- Who edited it, with old value, new value, and timestamp
- IP address of the acting user
- Signature time
- Download history

11. Invoice History Timeline

Every invoice exposes its own timeline view, distinct from the system-wide status lifecycle in Section 2:

Received → Extracted → BA Edited → Submitted → Approved → Signed → Finance Received → Completed

12. Analytics

- Monthly invoice volume
- Average approval time
- Invoice volume by vendor
- Extraction accuracy trend
- Most-rejected vendors/invoices
- Department-level approval delays

13. Storage

Original scope stored only the source PDF and the signed PDF. Revised scope stores the full artifact set needed for audit defensibility and future integrations:

- Original invoice (source file)
- AI-extracted JSON (pre-correction)
- BA-corrected JSON
- Generated Approval Document (PDF)
- Signed PDF
- Audit log entries

- Email metadata

14. Impact on Timeline and Budget

These additions materially expand MVP scope beyond the original 8–9 week estimate. Dashboards, the notification engine, search, expanded audit logging, RBAC, and analytics are each a meaningful build item on their own — not incremental polish on the original five modules.

Area	Original Scope	Revised Scope	Est. Added Effort
Status model	3 states	8-state lifecycle + timeline view	2–3 days
Duplicate detection	Future enhancement	MVP, blocking check	3–4 days
Dashboards	None	4 role-based dashboards	1.5–2 weeks
Notifications	Brief mention	Full engine incl. reminders	1 week
Search	None	Multi-field global search	3–4 days
Audit logs	Basic	IP, downloads, old/new value diffing	3–4 days
RBAC	Mentioned	5 roles incl. Read-Only Auditor	2–3 days
Analytics	None	6 report views	1 week
AI pipeline / validation	2-stage + confidence score	9-stage pipeline + 9 validation rules	1–1.5 weeks
Storage	PDF only	7-artifact structured storage	2–3 days

Revised estimate for this complete, single-phase build: 12–14 weeks (up from 8–9), with a proportional budget increase — indicatively AUD 42,000–56,000 for development effort, before infrastructure, AI/OCR/e-signature usage, and QA. Full breakdown by phase is in Section 19.

15. System Architecture

Nine independently deployable, stateless services communicating over REST, sequenced by a central orchestration layer that tracks invoice status through the full lifecycle (Section 2).

Service	Responsibility
Ingestion Service	Mailbox polling, attachment download, de-duplication of processed emails
Security Service	Virus/malware scanning of every attachment before it enters the pipeline
Extraction Service	OCR, classification, invoice detection, AI field/line-item extraction, confidence scoring
Validation Service	Executes all 9 validation rules and the duplicate-invoice check (Sections 1, 5)
Review Service	BA queue, correction interface, side-by-side source view
Approval Service	PDF generation, Department Head routing, comments, digital signature capture
Notification Service	All triggers and reminders in Section 8
Search & Analytics Service	Global search index (Section 9) and the report views in Section 12
Storage & Audit Service	Persists the 7-artifact set (Section 13) and every audit event (Section 10)

16. Database Schema (Core Tables)

Table	Key Fields
invoices	id, vendor, invoice_number, po_number, invoice_date, due_date, tax, total, currency, status, confidence_score
line_items	id, invoice_id (FK), description, quantity, unit_price, line_total
invoice_versions	id, invoice_id (FK), ai_extracted_json, ba_corrected_json, version, created_at
approvals	id, invoice_id (FK), department_head_id, comments, signature_ref, signed_at, decision
audit_log	id, invoice_id (FK), user_id, action, old_value, new_value, ip_address, timestamp
notifications	id, invoice_id (FK), recipient_role, trigger_type, sent_at, channel
users	id, name, email, role, department, is_active
duplicate_flags	id, invoice_id (FK), matched_invoice_id, status, resolved_by, resolved_at

Signed PDFs and source attachments are held in object storage; the database holds references and metadata only, per Section 13.

17. API Contracts

- /ingestion — mailbox status, intake records, de-dup checks
- /extraction — submit document, retrieve structured output + confidence scores
- /validation — run validation rules and duplicate check, return pass/fail with reasons
- /review — fetch pending invoices, submit BA corrections
- /approval — generate Approval Document, submit comments + signature
- /notifications — trigger and query notification history
- /search — multi-field query across invoices
- /dashboards — role-scoped metrics (BA, Department Head, Finance, Admin)
- /audit — query audit trail by invoice, user, or date range
- /storage — retrieve any of the 7 stored artifacts for a given invoice

18. Technology Stack

Layer	Stack
Backend	Python, FastAPI, pydantic
Database	PostgreSQL
Search	PostgreSQL full-text search (or OpenSearch if volume requires it)
Frontend	React + Tailwind, role-scoped dashboard views
Document Parsing	pypdf / pdfplumber, pytesseract (OCR fallback)
AI Extraction	LLM-based schema-constrained extraction API
Digital Signature	Certificate-based PDF signing library or client-approved e-signature API
Storage	Client-approved cloud storage, Australian region
Notifications	Email/SMTP or Microsoft Graph, with a scheduled job for 24h/48h reminders

Layer	Stack
Auth	OAuth2 / JWT, role claims for the 5 RBAC roles
DevOps	Docker, GitHub Actions (CI/CD)
Monitoring	Structured logging + APM/error tracker

19. Delivery Plan

Phase	Scope	Duration
Phase 0 — Discovery	Mailbox access, sample invoices, routing rules, signature method	1 week
Phase 1 — Ingestion & Security	Module A + malware scanning	1–1.5 weeks
Phase 2 — AI Pipeline	9-stage extraction pipeline + confidence scoring + validation rules	2–2.5 weeks
Phase 3 — Review & Duplicate Detection	BA correction UI, duplicate flag workflow	1–1.5 weeks
Phase 4 — Approval & Signature	PDF generation, Department Head workflow	1–1.5 weeks
Phase 5 — Notifications & Search	Full notification engine, global search	1–1.5 weeks
Phase 6 — Dashboards & Analytics	4 role dashboards, 6 analytics views	1.5–2 weeks
Phase 7 — Audit, RBAC & History	Expanded audit log, 5-role RBAC, invoice timeline	1 week
Phase 8 — Integration & Hardening	End-to-end testing, CI/CD, documentation	1 week

Total: 12–14 weeks. Recommended team: 1 Tech Lead, 2 Backend Developers, 1 Frontend Developer, 1 AI/ML Engineer (part-time), 1 QA Engineer (from Phase 3 onward).

20. Acceptance Criteria

- Every invoice in the sample set moves through all 8 lifecycle states with no manual database intervention
- Duplicate detection correctly flags 100% of intentionally duplicated sample invoices
- All 9 validation rules block submission when violated, with a clear reason shown to the BA
- Each of the 4 role dashboards shows live, correct figures against the sample data set
- Search returns correct results across all 7 supported fields
- Audit log captures every view, edit, signature, and download with correct old/new values and IP address
- All 5 RBAC roles are enforced — no role can access another role's actions
- Reminder notifications fire at 24h and 48h on an unsigned approval in the test environment
- Finance Team confirms the 7-artifact handoff package is sufficient for manual BMS entry with no follow-up requests